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Bulk electricity storage technologies for load-leveling operation – An economic assessment for the Austrian and German power market

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ABSTRACT

Fast growing shares of intermittent renewable electricity have raised the interest in the economic potential of electricity storage in Europe. In this paper we analyze the economic performance of bulk electricity storage (ES) technologies for load-leveling operation in the German and Austrian power market. We assess four different power storage technologies: Pumped hydro storage (PHS), adiabatic compressed air energy storage (AA-CAES), hydrogen storage (H_2) and methane storage (CH_4). We examine storage revenues by simulating optimal price arbitrage based on historic price data of the years 2007–2011.

The results indicate that revenues have decreased significantly during this time period and even the most efficient technology, the PHS, has fallen below the profitability margin. Alternative technologies proved to be inferior to the PHS as a day storage plant. There might be a field of application for H_2 and CH_4 as seasonal storage, if their costs can be reduced considerably.

The results reveal that, based on the past 5 years' development, incentives to invest in storage capacity are currently low. Given the uncertain mid- to long-term development of the power market, this could seriously delay the extension of storage capacities required for the integration of intermittent renewable electricity.

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1. Introduction

The European Union is currently driving an effort to increase the share of renewable electricity (RES-E) in order to reduce carbon emissions in the power sectors. This development has led to considerable shares of RES-E in countries like Germany and Austria, and a further increase is expected for the upcoming years [1]. The expected shares of fluctuating RES-E production will be a great challenge for the power system calling for more flexibility of back-up power plants to cover the residual load and higher transmission capacities to balance supply and demand in wider areas.

Energy storage is considered a key component to face these challenges. Previous work has shown that they can help to balance fluctuations in renewable production both on a local and on a regional level. Refs. [2–4] demonstrate the technical and economic benefits of storage use in isolated power systems. Ref. [5] shows the benefits of combining wind farms with power storage. Ref. [6] indicates that power storage increases the overall social welfare in the power market and [7] find that power storage leads to benefits in a power system with high wind penetration. Ref. [8] demonstrates how electric storage plants can help to reduce transmission costs in a power system with wind generation.

Ref. [9] find that storage plants are more viable than thermal power plants as a flexibility option until a certain installed capacity. This is in line with the results found by [10,11] who analyzed optimal economic storage capacity for the German power system.

While economic benefits of power storage seem evident on a power system level, their economic performance from an investor perspective remains uncertain. Ref. [7] shows that the profitability of a compressed air energy storage (CAES) remains negative even though it brings about economic benefits to a power system. Ref. [12] analyzes the current and the future economic potential of a CAES in the Nordic "Nord Pool" market. Ref. [13] performs a model-based economic assessment of power storage plants in different European countries based on historic price data. He finds that internal rates of return are negative for most of the analyzed technologies and countries in the analyzed time period (2003–2004). The analysis of [14] demonstrates that feed-in of intermittent RES-E has reduced revenues of PHS plants in the German and Austrian power market.

In this paper we analyze the economic performance of different bulk storage technologies in load-leveling operation. We have

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selected four technologies that are currently considered most relevant for large-scale load-leveling in the future power system:

- Pumped hydro storage (PHS)
- Adiabatic compressed air energy storage (AA-CAES)
- Hydrogen storage (H_2)
- Methane storage (CH_4)

In the economic assessment we analyze these technologies in a power market environment. We selected the German power market because of its fast growing capacity of intermittent RES-E capacities. We used clearing price data of the EXAA (Energy Exchange Austria), which are almost perfectly correlated with EEX (European Energy Exchange) clearing prices, because of sufficient transmission capacities between Austria and Germany.

Based on data of the years 2007–2011 we examine the yearly revenues of the four power storage technologies in load leveling operation. We use a model that simulates optimal storage operation we evaluate the maximum yearly revenues and compare it with estimated capital costs.

With the results we want to provide an insight in the current economic performance of power storage as well as the development in the analyzed time period. They may aid to understand power storage operation in a market environment and particularly the specific differences of storage technologies. This is especially relevant with respect to storage options that are currently in a development stage (e.g. power-to-gas technologies) and the future economic potential is still uncertain. With its focus on the common Austrian and German power market, the results provide an insight in the effects of increasing intermittent RES-E capacity on the storage operation and revenues.

The remainder of the paper is structured as follows:

Section 2 gives an overview on technical and economic parameters of the analyzed ES technologies. Section 3 explains the methodology of the economic assessment and the storage model. Section 4 presents the relevant economic framework conditions and input data that are used for the assessment. In Section 5 the results of the assessment are presented and in Section 6 the main findings are discussed. Section 7 draws conclusions.

2. Storage technologies

In this section we will present the four analyzed power storage technologies. We selected four technologies that are currently considered most relevant to facilitate the integration of intermittent RES-E in the central European Power market. In addition to pumped hydro storage (PHS), which is used as the benchmark, we looked at three alternative options that could fulfill similar tasks in the future power system. All these technologies can be applied as large-scale central storage technologies, but in fact their requirements with respect to site conditions are different. While PHS rely on certain geographic conditions (mountainous or at least hilly), the CAES and H_2 storage need specific geological underground formations, whereas the CH_4 needs suitable natural gas infrastructure near the site.

We will now briefly present the functioning of these technologies and look into their technical and economic parameters.

2.1. Pumped hydro storage (PHS)

PHS (see Fig. 1) is a mature technology that is widely used as an electric storage for different functions in the power system and widely applied in Europe [15,16]. The ability of PHS plants to integrate intermittent wind sources has been proved in various studies (e.g. [3–5,17,18]). Also there are several studies that underline the economic benefits of PHS in the power system as well as for investors (see [11,13,19]).

A PHS is typically applied for load-leveling, but they can also provide control reserve due to their relatively short response time.

Cycle efficiency of PHS ranges from 65% to 87% depending on the technological status of the plant [20–22]. For our analysis we assume a newly build plant with state of the art technology and a cycle efficiency of 85%.

Power rates of central PHS plants usually range from a couple of hundred to up to 1500 MW (see [15]). Investment costs of PHS are usually expressed as cost per unit of power installed with values ranging from 500 €/kW to 1500 €/kW [21]. Refs. [15,16] show that investment costs of PHS build in the past actually ranged from 500 €/kW to over 2000 €/kW. Values for storage capacity specific investment costs in literature range from 5 to 100 €/kW h [20,21]. To estimate investment cost of a newly built PHS we chose a two-dimensional approach with a power-rate-dependent and a storage-capacity-dependent cost component:

$$IC_{\text{total}} = ic_{\text{power}} \cdot C_{\text{power}} + ic_{\text{capacity}} \cdot C_{\text{capacity}} \quad (1)$$

where ic_{power} and ic_{capacity} are the power-rate- and storage-capacity-related investment costs, which are used to estimate investment cost of a PHS with a power rate C_{power} and an energy storage capacity C_{capacity}. Comparison with existing projects showed that 500 €/kW and 30 €/kW h provide a good approximation for a plant at a favorable site. The facts that such sites are getting increasingly scarce in Germany and Austria and that suitable sites are often in remote, mountainous areas are critical handicaps to this technology.

2.2. Adiabatic compressed air energy storage (AA-CAES)

Compressed air energy storage (CAES) plants use the potential energy of compressed air to store electricity. The electricity is used to drive compressors that compress the air at high pressure to store it either in pressure vessels or in geological formations (e.g. salt caverns). When released, the air is used to drive turbines that generate electricity. There are two types of compressed air ES plants: diabatic (CAES) and (advanced) adiabatic plants (AA-CAES). Globally there are currently only two diabatic CAES plants in operation [23] while AA-CAES are still in a conceptual phase [24]. Large-scale CAES rely on the availability of geological formations (e.g. salt cavern) that are suitable in size and durability for storage operation. Potential sites where a large-scale CAES can be installed are therefore limited.

In diabatic plants natural gas has to be co-fired while discharging to prevent the turbine from freezing during expansion of the air. In AA-CAES the heat accumulated during compression is reused for this purpose. Hence, an AA-CAES consists of a compressed air storage and a thermal energy storage system, making the technology more complex and costly but also more efficient (see Fig. 2). While CAES have electric cycle efficiencies of 40–50% [23] AA-CAES are projected to reach efficiencies between 60% to up to 80% [23,25,26].

In the analysis we only consider an AA-CAES since it offers higher efficiency than the diabatic CAES and it does not need fuels to be co-fired when discharged. Since AA-CAES are still in the planning and demonstration status (e.g. the ADELE Project in Germany), we estimated investment cost based on data found on CAES and thermal storages. Similar to the PHS values for investment costs of CAES found in literature are either defined as power-rate dependent or storage-capacity-dependent costs. Values for storage-capacity-dependent costs range from 5 to 100 €/kW h [20,22]. Values for power-factor-depending investment costs in literature are ranging from 400 to 800 €/kW [7,12,21,23,27]. For our cost estimation we used an approach that considers both cost components. Capacity

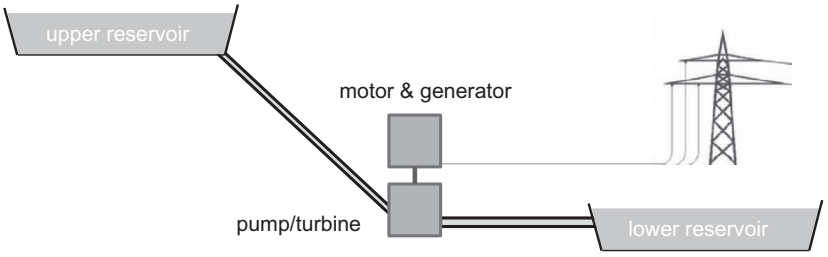


Fig. 1. Schematic diagram of a PHS.

related costs for AA-CAES systems should be in the range of CAES plants since the main components, mainly compressor and expander, are similar. Therefore, we set them at 600 €/kW, which is in line with the costs of CAES found in literature. Storage capacity-dependent costs will be higher for AA-CAES than for CAES systems since the first requires a thermal storage in addition to the cavern for the compressed air. Thus, we used capacity-dependent costs of 70 €/kW h representing storage for compressed air and heat.

2.3. Hydrogen storage (H_2)

In hydrogen storage systems electricity is converted into hydrogen via electrolysis. The hydrogen is then stored either in pressure vessels or, for central large-scale plants, in suitable geological formations (e.g. salt or rock caverns). To reconvert the stored energy into electricity, hydrogen can either be fired in thermal power plants or reconverted in a fuel cell. For large-scale storage on power system level this would be done either in a single cycle gas turbine (GT) or in a combined cycle gas turbine plant (CCGT). In this case a storage cycle efficiency of 33–40% could be achieved (see Fig. 3) based on component efficiencies derived from [28] (see Table 1). Since only demonstration plants for hydrogen storage have been realized so far (e.g. the pilot plant in Falkenhagen, Germany), investment costs have to be estimated based on the system components. Thereby, we used the figures provided by [27,29], where current costs of an electrolyzer are estimated to be 1000 €/kW. With scale effects [27] see a short- to medium-term reduction potential down to 500 €/kW. The thermal power plant investment costs depend on the technology applied and range from 300 €/kW for GT to 600 €/kW for CCGT plants in Europe (cf. [30–32]). For the hydrogen storage cavern we used 0.5 €/kW h according to [27] (see Table 1).

2.4. Renewable methane storage (CH_4)

Another approach to store electricity is the conversion into methane. Hereby, electricity is converted into hydrogen first and then turned into methane by adding CO_2. The methane can be fed in the natural gas grid, respectively natural gas storage facilities can be used for storage. Therefore, the plant requires an appropriate access to the natural gas grid near the site. The reconversion to electricity can be done by conventional natural gas fired power plants (e.g. GT or CCGT plants). The upside of this concept is the fact that with methane as a storage medium the existing storage and distribution capacity of the natural gas infrastructure could be used. With its high capacities for both storage and transportation this could alleviate some major problems associated with intermittent renewable electricity supply. The downside of the concept is its poor storage efficiency. We estimated the cycle efficiency based on the values provided by [28] with additional losses for the compression to 200 bar pressure needed for transport pipelines and storage (see Fig. 4 and Table 1). According to [28] the methanator reaches an efficiency of 80% which is in line with the value provided by [33]. This leads to a cycle efficiency between 27% and 32% depending on the efficiency of the power plant.

Investment costs of a large facility are estimated based on the costs of components involved. Hereby, the methanator is an uncertainty factor since no data on that component could be found. Therefore, we estimate investment costs to be similar to an electrolyzer (see Table 1).

3. Economic assessment

In the assessment we analyze technologies from an investor's perspective. This means that plants are operated to maximize their own profit and not to improve the efficiency of the power system or power market unless contractually set. This approach corresponds to the given framework conditions in liberalized power markets, where ES plants are privately owned and sell their services on the market. In this section an insight in the economic operation of electric storages is given.

3.1. Technology parameters

In Table 2 the parameters that are used for the further assessment as well as some of their technical characteristics are

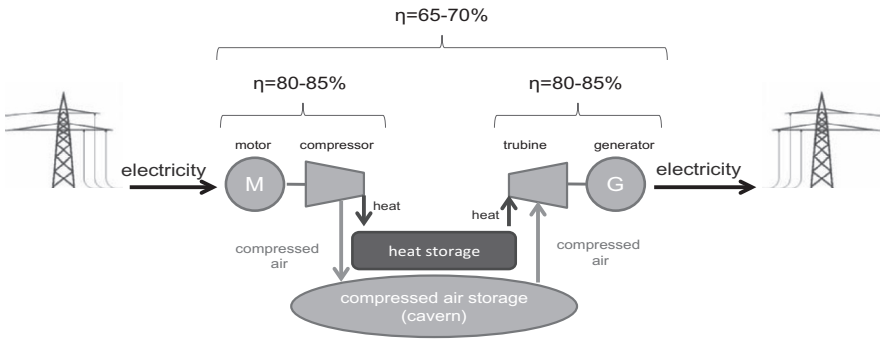


Fig. 2. Schematic diagram of adiabatic compressed air energy storage (AA-CAES).

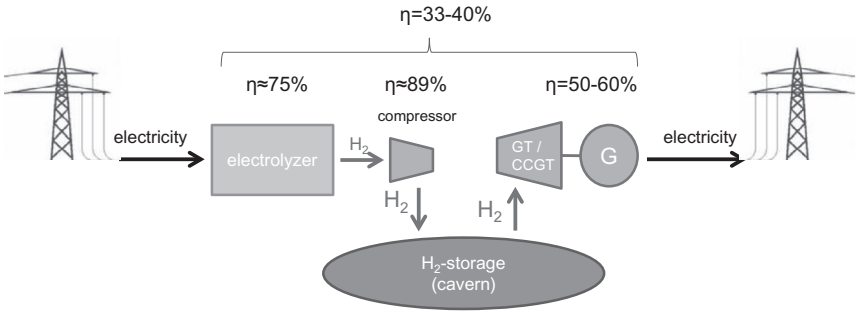


Fig. 3. Schematic diagram of a H_2-storage system.

Table 1
Efficiency and investment costs of hydrogen and methane storage components.

	Efficiency (%)	Investment costs (€/kW)	Sources
Hydrogen storage (H_2)			
Electrolyzer	75	500	[27,28]
Compressor	89		[28]
Thermal power plant (GT or CCGT)	50–60	300–600	[30–32]
Total	33–40	800–1100	
Methane Storage (CH_4)			
Electrolyzer	75	500	[27,28]
Methanator	80	500	[27,28]
Compressor	97		[28]
Thermal power plant (GT or CCGT)	50–60	300–600	[30–32]
Total	29–35	1300–1600	

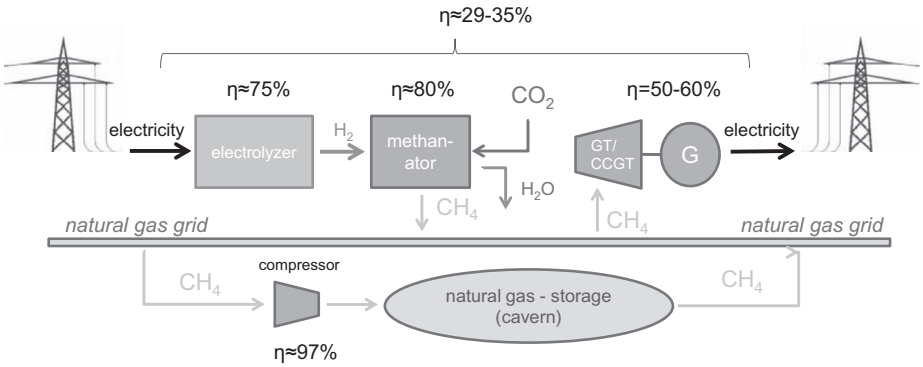


Fig. 4. Schematic diagram of a renewable methane-storage system (CH_4).

Table 2
Storage parameters used in the analysis.

			Pumped hydro storage (PHS)	Adiabatic compr. air energy storage (AA-CAES)	Hydrogen storage (H_2)	Methane storage (CH_4)
Efficiency	Charging	(%)	92	84	67	58
	Discharging	(%)	92	84	60	60
Investment costs						
Capacity specific		(€/kW)	500	600	1000	1500
Energy specific		(€/kW h)	30	70	0.5	0.15
Depreciation time		(years)	25	20	20	20
O&M costs		(€/kW/year)	4	20	20	20

summarized. For the selected parameters we used values that imply technological progress especially for those technologies that are currently in an experimental or demonstration phase.

3.2. Costs and revenues of storage operation

The profit of the ES plant G is defined by the revenues from optimal peak/off-peak arbitrage operation R_{p-o}, capital costs C_c and maintenance costs $C_{o\&m}$.

$$G = R_{p-o} - C_c - C_{o\&m} \quad (2)$$

G – profit (€/year); R_{p-o} – revenues from peak/off-peak spread (€/year); C_c – capital costs (€/year); $C_{o\&m}$ – operation and maintenance costs (€/year).

As indicated by Eq. (2) there are different types of revenues a storage plant can generate. With energy arbitrage, the ES uses the spread between peak and off-peak hours in the spot market price to generate revenues. Thereby, revenues depend on the course of electricity prices during the time period and the efficiency of the storage (see Eq. (3)).

$$R_{p-o} = f(p_{e(t)}, \eta_s) \quad (3)$$

p_e – Electricity spot market price (€/MW h); η_s – Storage cycle efficiency (%).

Thereby, the economic criterion for the storage to operate is that its efficiency has to be greater than the ratio of acquisition and sales price of electricity (see Eq. (4)) (cf. [19]).

$$\eta_s \geqslant \frac{p_{in}}{p_{out}} \quad (4)$$

p_{in} – Electricity acquisition price (€/MW h); p_{out} – electricity sales price (€/MW h).

Instead of or in addition to energy arbitrage the storage can generate revenues from ancillary services, for example by providing control reserve. These types of revenues will be discussed in Section 4.

Capital costs are calculated using the capital recovery factor defined by the technology specific depreciation time (see Table 2) and an interest rate of 7%.

$$CC = IC \cdot CRF \quad (5)$$

$$CRF = \frac{r \cdot (1+r)^{DT}}{(1+r)^{DT} - 1} \quad (6)$$

CC – capital costs (€/MW h/year); IC – investment costs (€/MW h); CRF – capital recovery factor; r – interest rate (%); DT – depreciation time (years).

To determine capital costs of the plants we used the approach presented in Section 2, which considers cost components that are dependent on the power rate and on the storage capacity.

Optimal storage operation is simulated using a linear optimization model implemented in General Algebraic Modeling System (GAMS). The model determines the optimal storage operation schedule in order to maximize revenues from peak/off-peak arbitrage using historic yearly electricity price data.

The objective function is maximization of yearly profit:

$$\max[R_{p-o}] \text{ with } R_{p-o} = \sum_t (p_{(t)} - c_l) \cdot P_{out} - (p_{(t)} + c_l + c_g) \cdot P_{in} \quad (7)$$

p – Electricity price (€/MW h); c_l – grid loss fees (€/MW h); c_g – variable grid connection fees (€/MW h); P_{in} – electric storage input (MW); P_{out} – electric storage output (MW).

State of charge of the storage is defined as follows:

$$C_{(t)} = C_{(t-1)} + P_{in} \cdot \eta_{in} - P_{out} \cdot \eta_{out}^{-1} \text{ with} \quad (8)$$

$$0 \leqslant C \leqslant C_{max} \quad (9)$$

$$0 \leqslant P_{in} \leqslant P_{max} \quad (10)$$

$$0 \leqslant P_{out} \leqslant P_{max} \quad (11)$$

C – state of charge (MW h); η_{in} – charging efficiency (%); η_{out} – discharging efficiency (%); C_{max} – state of charge (MW h); P_{max} – maximum electric input or output (MW).

The model uses historic price data to simulate storage operation and determine yearly revenues. This means that it has perfect foresight, thus generating the maximum achievable annual profit. In practice plant operators have to rely on their forecasting techniques and their results will be some percent below the optimal values.

We neglected the effect that an additional power plant would have on spot market prices. Additional storage capacity tends to level spot market prices and thereby reduces revenues of all storage plant [10,19]. Accordingly, a new plant, as it is assumed in our analysis, would affect the course of spot market prices.

4. Economic framework conditions and input data

In this section we provide a brief insight in the economic framework conditions for ES in the Austrian–German power market. This background information should allow for a better understanding of the results that will be presented in Section 5. We start with the input data of the analysis and proceed with an overview of the structure of power generation in the analyzed market area.

4.1. Austrian–German electricity market area

Due to better data availability we used historic electricity prices of the Energy Exchange Austria (EXAA) electricity spot market for the years 2007–2011. Because of the high transmission capacity between the Austria and Germany the two countries form a price zone and EXAA electricity prices are almost perfectly correlated to the (EEX) day-ahead prices.

For a better understanding we give a brief overview on the current (status 2011) power generation capacities in Germany and Austria which are the most relevant countries for the EEX and the EXAA spot market prices. Fig. 5 illustrates the development of power generation capacities in the years 2007–2011. It shows that base-load capacities mainly comprise of coal, lignite, nuclear and run-of-river hydro plants. While base load capacity has remained more or less constant, there has been considerable increase in gas-fired power plant capacities. However, the most significant development throughout the analyzed time period has definitely been the increase of intermittent photovoltaic (PV) and wind generation capacities. Wind has grown from 23 GW to 29 GW and PV has leaped from 4 GW in 2007 to 16 GW in 2011. Hydro storage capacity in these two countries has remained more or less constant throughout the last years with around 15 GW of turbine power and 11 GW pump power (data sources: [34–37]).

4.2. Control reserve and intraday market

In some European countries provision of power reserve has been an important revenue stream for electricity storage systems in the past years (see [13]). With an increasing share of intermittent renewable electricity production the relevance of control reserve is expected to increase in the future. On the other hand, consolidation of control energy markets over control areas will lead to higher efficiency in control reserve use and consequently

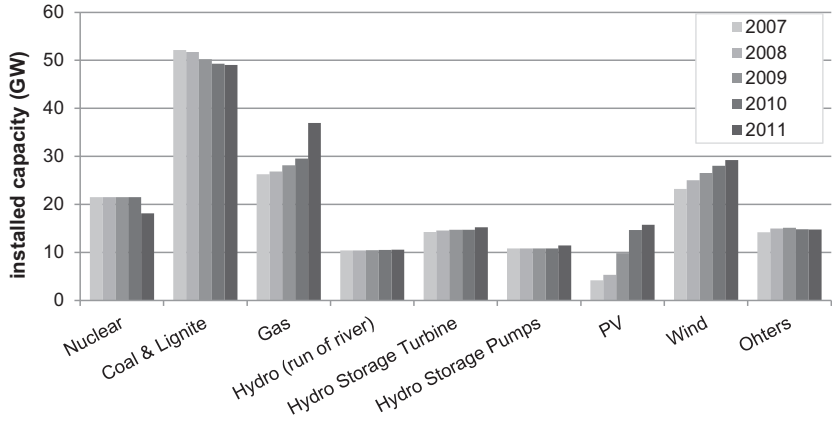


Fig. 5. Generation capacity in the EEX/EXAA market area (data sources: [34]–[37]).

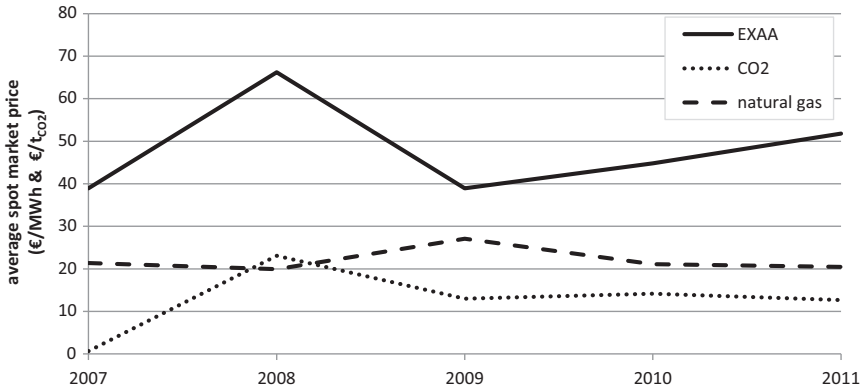


Fig. 6. Average yearly spot market prices of electricity (EXAA), natural gas and CO_2.

to lower revenues for control reserve providers (see [38–40]). Demand for control reserve is currently defined and tendered by various system operators in Europe with differences in the tendering process as well as average prices for control reserve (see [41,42]). For a power plant potential revenues from power reserve depend on total demand in the control area, amount and cost structure of the competing capacity, as well as the probability of reserved capacity to be actually called. Also bidding on the reserve market always implies the opportunity costs of not being able to participate on spot markets.

All these factors bring about uncertainty and do not allow a general estimation of the revenue potential of new power plants on control reserve markets in the EXAA and EEX market area. Therefore, we did not consider revenues from control reserve in this analysis and focus on revenues from load-leveling. However, it has to be kept in mind that provision of control reserve might improve the economic performance especially for highly flexible plants like PHS.

Also the importance of intraday markets is supposed to grow with the ongoing integration of large amounts of intermittent RES-E. Already now, the electricity prices at the German EPEX SPOT intraday market show high differences between base and peak-load periods allowing bulk ES technologies to profit from arbitrage operation. Today volumes traded on the intraday market are still low compared to the day-ahead market but its relevance has increased considerably in the past years and is expected to continue to do so in the upcoming years. This will make them an interesting alternative revenue stream for power storage plants.

5. Results

The model described in Section 3 is used to assess storage technologies in a market environment. We simulate operation for the four selected technologies. In a first step we analyzed the optimal layout of storage capacity from an economic point of view. Fig. 7 illustrates yearly revenues of storage plants based on 2009 spot-market prices as well as estimated capital costs, both expressed as a function of storage capacity. Thereby, capital costs are estimated based on the approach presented in Section 2.

The results show that in 2009 revenues of all analyzed technologies have fallen below their estimated capital costs. While revenues of the PHS reach close to capital costs, they are well below their respective costs for all other technologies. From this figure we can derive that optimal layout of a PHS storage capacity would be with a storage capacity to power rating ratio of 7–8. This means that the optimal way to operate a PHS plant is as a day-storage. The same applies for the AA-CAES with a slightly smaller ratio of 4–5. For H_2 and CH_4 yearly revenues are far below their estimated capital costs. However, with their low capacity-dependent investment costs they could operate with much higher ratios and potentially serve as seasonal storage plants. A closer look on this option will be taken later in this section.

First we want to take a closer look on the day storage operation using an energy-to-power-ratio of 7. With these parameters we compared the yearly revenues of the four technologies for the years 2007–2011 (see Fig. 8). Generally, it shows that storage revenues are highly sensitive to changes in the economic framework conditions. Between 2009 and 2011 calculated revenues of the

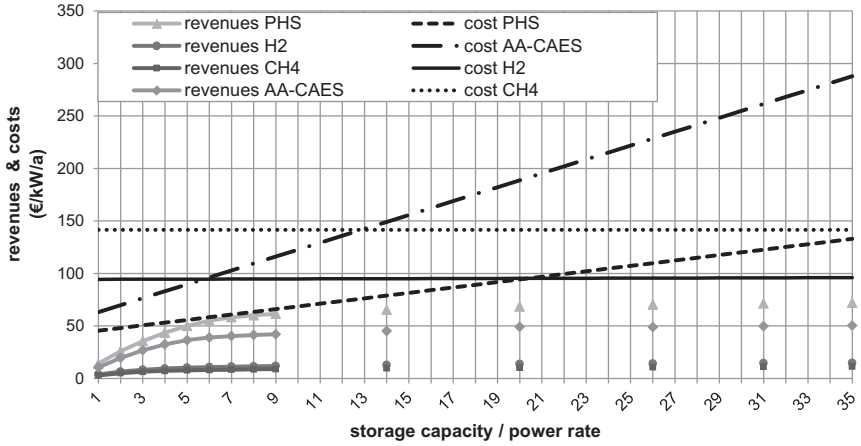


Fig. 7. Revenues and capital costs of ES technologies as function of the storage capacity based on electricity prices 2009.

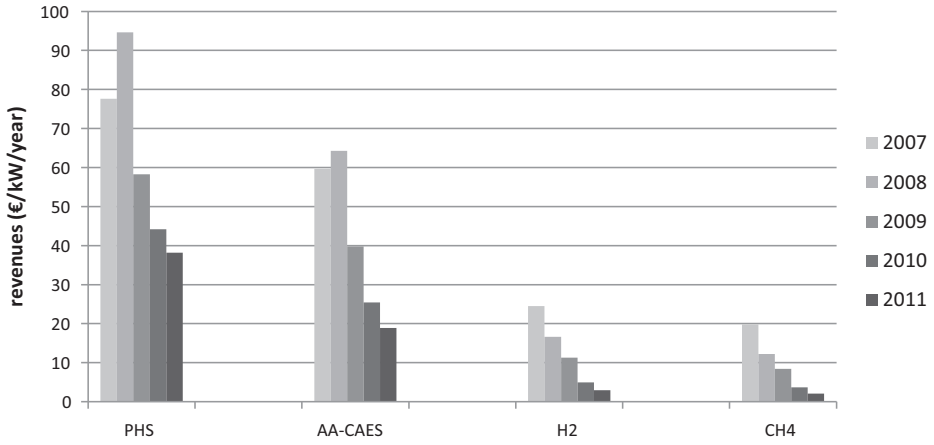


Fig. 8. Yearly revenues from storage operation for day storage operation with different technologies.

PHS have dropped by 60%, those of the AA-CAES by around 70% and those of the H_2 and CH_4 by more than 80%. This indicates that technologies with lower cycle efficiency are much more sensitive to changes in spot market prices.

To understand these differences in revenues of the different technologies a closer look on their daily operation schedule has to be taken. Figs. 9 and 10 compare storage operation of analyzed technologies for 300 MW plants in a December (December 14–20, 2009) and a June week (June 15–21, 2009). The technology-specific differences are evident. The PHS and the AA-CAES show a very similar profile, but even the 10% difference in cycle efficiency affects storage operation. The PHS has more operation hours and uses the total storage capacity also during summer weeks. For the H_2 and the CH_4 the operation schedule is fundamentally different. Because of their low efficiency they can hardly use price spreads for daily cycles. They rather operate in week storage schedule leading to significantly lower revenues. This is underlined by the yearly total power input of the four technologies in the five analyzed years

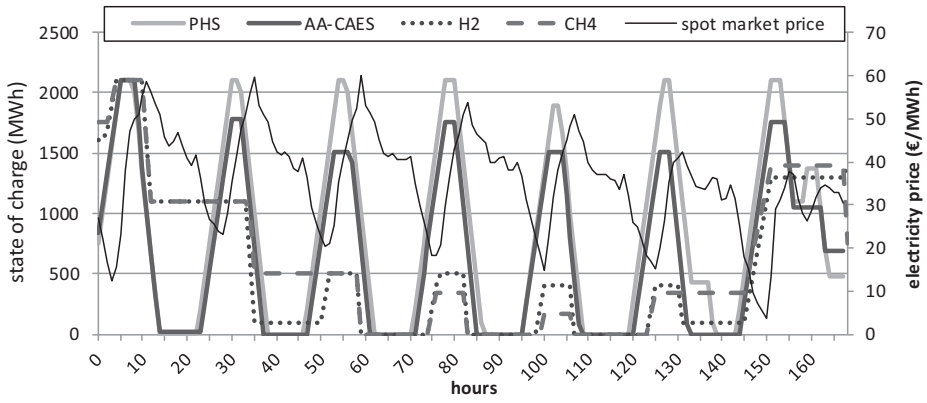


Fig. 9. Weekly storage operation of different technologies (June 15–21, 2009, (power rate = 300 MW, storage capacity = 2100 MW h_{el_out}).

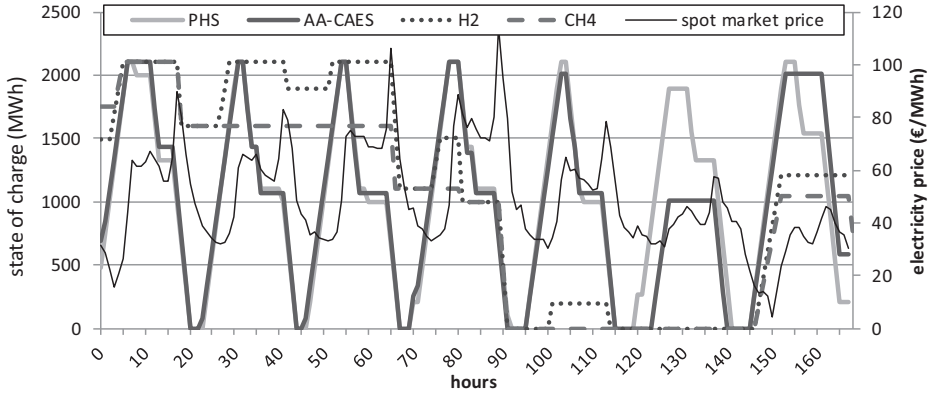


Fig. 10. Weekly storage operation of different technologies (December 14–20, 2009, power rate = 300 MW, storage capacity = 2100 MW h_{el_out}).

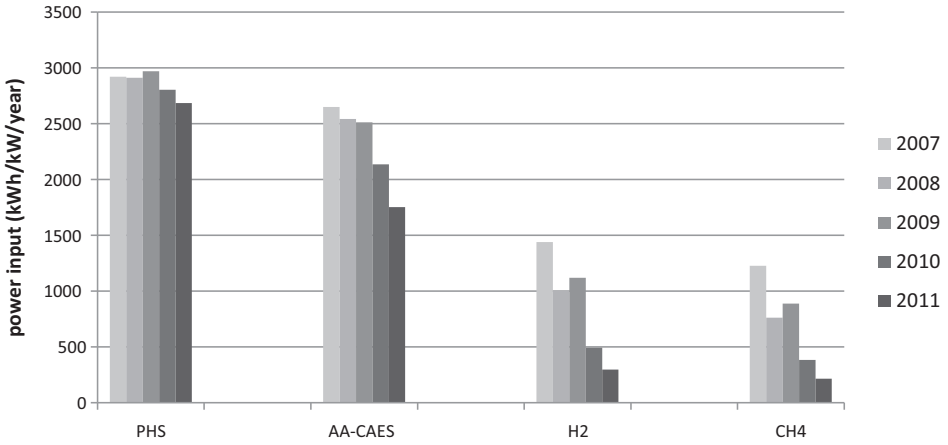


Fig. 11. Yearly power input in day storage operation for different technologies (power rate = 300 MW, storage capacity = 2100 MW h_{el_out}).

illustrated in Fig. 11. It indicates that technologies with high cycle efficiencies are less sensitive to changes in spot market prices than alternatives with lower efficiency. In case of the H_2 and CH_4 the effect of the price changes are tremendous, with a reduction of 80% between 2007 and 2011.

In a next step we analyzed seasonal storage operation. Balancing the discontinuous availability of renewable electricity throughout the season asks for huge storage capacity. This means that, technologies with low capacity-dependent investment costs are preferred for this task. Due to their technological characteristics H_2 and CH_4 storages are particularly suitable for seasonal storage operation. Therefore, we also simulated this type of operation assuming that the plant with defined power rate (300 MW in our case) can use as much storage capacity as required in order to maximize its yearly revenue. In addition to the H_2 and the CH_4 storage we also simulated a PHS for seasonal operation. It is evident that, from an economic perspective, a newly built PHS would not be sized for seasonal storage operation. The comparison should just point out the technology-specific differences.

Fig. 12 shows the specific storage capacity used in seasonal storage operation for the three analyzed technologies based on prices of the year 2009. It indicates that all technologies follow the same

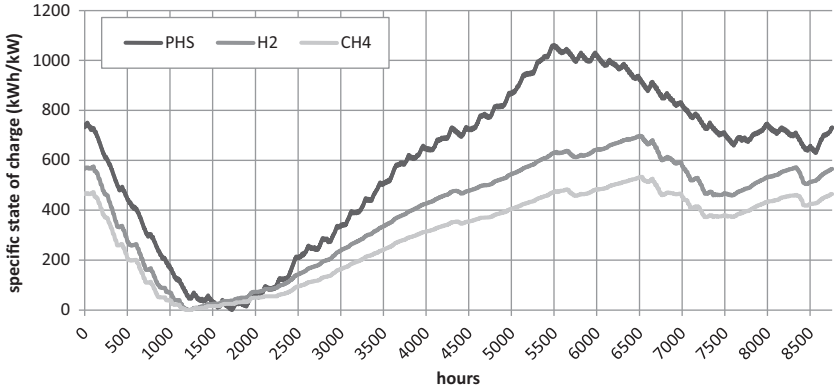


Fig. 12. State of charge of storage plants in seasonal operation (price data 2009).

strategy: charging the storage in spring and summer and discharging it in autumn and winter. However, the extent of storage capacity used differs between technologies. The higher the efficiency of the storage plant the higher the capacity used. There is also a difference in the weekly and daily cycles. While the PHS shows distinctive weekly and daily cycles in addition to the seasonal cycle, these are much less significant for H_2 and CH_4. This is also reflected in the yearly power input of the storage technologies depicted in Fig. 14. Similar to the day storage plants power input is far lower for H_2 and CH_4, and it is also much more sensitive to changes in the spot market prices, as the decline between 2007 and 2011 indicates.

Yearly revenues depicted in Fig. 13 show a similar trend of decreasing revenues for H_2 and CH_4. However, there is a little discrepancy for the PHS, where revenues showed much more sensitive to price-changes than the power input (cf. Figs. 13 and 14). In the year 2008 revenues even exhibit an opposite trend compared to the power input and compared to other technologies. These figures clearly point out the differences between the analyzed storage technologies from an economic perspective and indicate the significant impact of storage efficiency on their performance.

The derived results for both daily and seasonal storage operation are summarized in Table A-1 in the appendix. In addition to revenues and power input we also derived the economically feasible investment costs based on the assumption made in Sections 2 and 3. According to these figures the maximum investment costs of H_2 and CH_4 storage range from 51 to 581 €/kW and 33 to 463 €/kW depending on the analyzed years. Even in the best year (2007) this would cover only 58% (H_2) and 31% (CH_4) respectively of estimated capital costs.

6. Discussion

6.1. Findings on storage technologies

The results of the assessment show that, taking the years 2007–2011 as a basis, only the PHS could be an interesting investment for load-leveling provision in the Austrian-German power market. However, even here storage revenues have been declining during this time. For day-storage operation the AA-CAES reaches closest and might become an alternative to the PHS plants if investment costs can be proved and even further reduced for this technology.

H_2 and CH_4 are often considered as future options to balance demand and intermittent supply in longer time horizons. However, our results indicate that their capabilities in this function are limited in the current market environment and at their state of technology. Due to their low cycle efficiency they require considerable price spreads to operate. These can only be reached, if at all, for few hours per day. Hence, their contribution to intraday load-leveling in today's power system would be little.

The results show that daily cycles are the most important source of storage revenues and also account for the main part of the energy flow through the storage plant. Consequently, the optimal economic layout of a storage plant would be for day-storage cycles with a ratio of capacity to power-rate of 7–8. However, technologies with very low, respectively close to zero capacity-dependent capital costs, such as CH_4 and H_2, could be operated as a storage plant with almost unconstrained storage capacity. Under this condition they would increase their revenues by 50% (2011) to 120% (2007) compared to daily storage operation with constrained capacity (day-storage). In this case power flow would increase by 29% (2011) to 91% (2007). This indicates that these power storage technologies can contribute to inter-seasonal balancing of power-supply.

6.2. Findings on storage economy

The results show that there has been a general decline in storage revenues in the analyzed time period. This is an indicator for either decreasing demand for load-leveling or increasing supply to cover demand. There are various potential drivers that might have affected this development in the German–Austrian power market, which we will address one by one:

6.2.1. Base-load/off-peak generation capacity

Since storage plants use off-peak hours to buy electricity and fill their reservoirs, changes in base-load capacity have an effect on storage revenues. As illustrated in Section 4, there has been no significant change in the structure of base-load capacities in the analyzed years that might have triggered this development. With the phase-out of German nuclear plants in the upcoming years and decades increasing off-peak prices can be expected, which would have a negative effect on storage revenues. On the other hand the extension of well-connected wind capacities will compensate this price increase. [43] finds that the availability and the share of nuclear power plants and their phase-out "speed" are of minor importance for electricity prices in Germany.

6.2.2. Flexible generation capacity

Taking into account the development of the Austrian and German generation capacities in the years 2007–2011, we see that there has been an increase in natural gas-fired production capacity. New gas plants are usually highly efficient and flexible mid- to

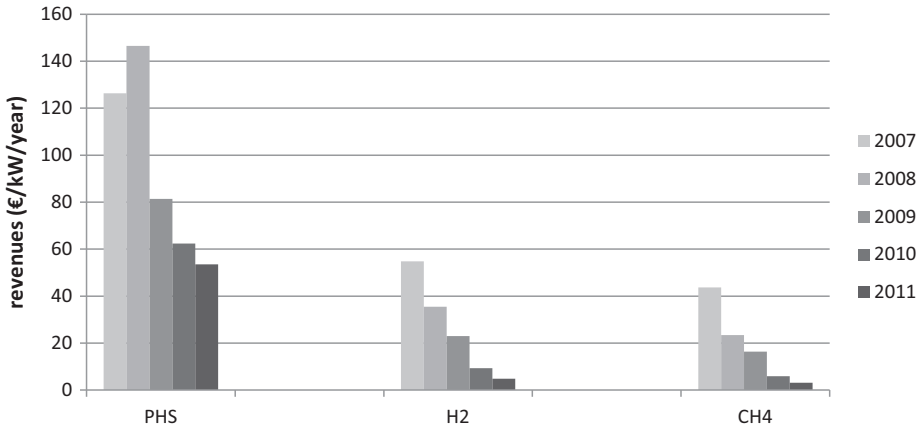


Fig. 13. Yearly revenues for seasonal storage operation with different technologies.

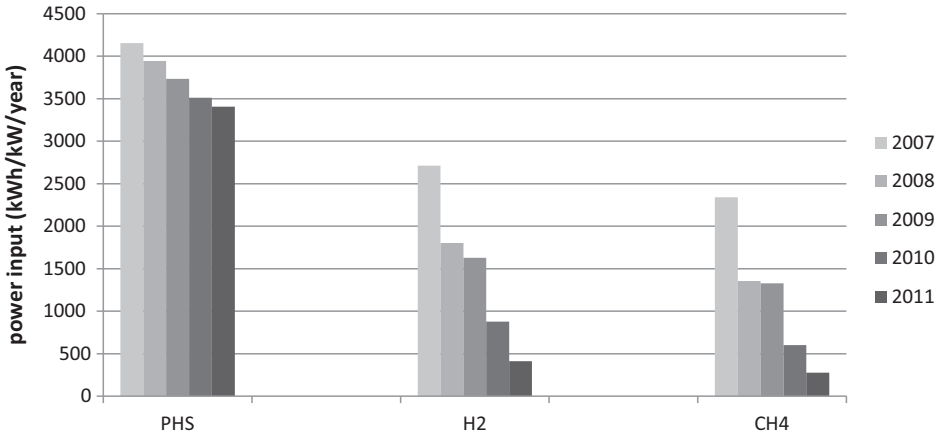


Fig. 14. Yearly power input in seasonal storage operation for different technologies.

peak-load plants that lower the slope of the supply curve. This leads to a decline of peak-load prices and consequently to a reduction of power storage revenues. Hence, extension of natural gas-based production capacity might have been one driver of the decreasing revenues in the past years.

Flexible generation capacity, especially natural gas based, is expected to further increase in the years and decades to come, which will put further economic pressure on storage plants (cf. [37]).

6.2.3. Storage capacity

Storage plants affect spot market prices in two ways. Through buying electricity they increase prices at off-peak hours and through selling they lower the prices at peak hours. Consequently, an extension of storage capacity levels the course of prices which lowers profits for all storage plants. In the analyzed time period total storage capacity has not been extended significantly, which indicates that this factor cannot be made responsible for the decrease in storage revenues. Model based analyses for the German power market have shown that the storage plants that are currently under construction and in the planning process will have an effect on power prices as soon as they go on line (see [10,19]). However, many of these projects have been postponed or cancelled due to uncertain economic prospects.

6.2.4. Carbon dioxide prices

Carbon dioxide prices have a strong impact on electricity market prices. They affect prices of both base-load and peak-load plants. Fig. 6 indicates the correlation of CO_2 certificate prices and average spot market prices in the years 2007–2011. It is evident that average prices do not allow a reliable judgment of storage revenues since the intraday development has to be taken into account. In a long run an increase in CO_2 prices is expected or at least desired by policy makers in order to foster low carbon generation technologies in the market. In a short run higher CO_2 prices could reduce average intraday price-spreads in the German power market (i.e. "compress" the merit-order curve), since they lead to a stronger increase in electricity generation costs of carbon intense base-load technologies such as coal and lignite, compared to the natural gas-based peak plants (cf. [44]). However, in the long-run higher CO_2 prices will lead to higher marginal costs of typical lignite- and carbon-based base load plants and to growing renewables capacities. This will drive parts of the current base- and medium load capacities out of the market and steepen the merit-order curve and thereby will have a positive effect on revenues from load leveling operation.

6.2.5. Natural gas prices

By affecting marginal costs of peak-load plants natural gas prices also affect storage revenues. Natural gas prices used to be linked to the oil price, which made them relatively volatile compared to coal or lignite prices. In the analyzed time frame natural gas prices remained relatively stable. This suggests that they are not responsible for the decline of storage revenues in this period.

6.2.6. Wind power

As an intermittent power source wind power strongly affects energy prices. Empirical analysis have shown that they lead to higher volatility of prices potentially leading to extreme price events (cf. [45–47]). With respect to power storage it is not clear whether this has positive or a negative effect on revenues throughout the year. High wind feed-in leads to a reduction of spot market prices or even negative prices in extreme cases. These periods could be used by the storage to buy electricity at low rates. However, wind can also reduce peak prices, which then reduce the revenues from the power sale.

Another problem of wind energy is its intermittent availability in cycles that are often too long to be utilized by typical storage plants, which are designed for day-storage operation. Our results show that most of the power flows and revenues of the storage plants are generated in daily storage cycles where wind power has less effect. Therefore, the effect of wind power capacity on load leveling revenues of day-storage plants should be insignificant.

6.2.7. Photovoltaic (PV)

As an intermittent power source PV power affects daily spot market prices. With their feed-in they lead to lower prices around midday where solar radiation is strongest. This means that they reduce prices exactly when the demand and consequently the power market prices used to be high. In summer this effect leads to a considerable reduction of daily midday price-peaks, which leads to a reduction of storage plant revenues (cf. [14]). Given the massive extension of PV capacity in the analyzed timeframe from 4 GW to 16 GW it is likely that this has been a main driver of the reduction in yearly storage revenues.

7. Conclusions

This paper provides an economic assessment of different bulk electricity storage technologies for load-leveling operation on the example of the Austrian-German power market. The results show that within the currently available technologies, the PHS is still the best option from an economic perspective. Due to its high

storage cycle efficiency it generates the highest revenues, while capital costs are in the range of alternative technologies. However, there is an uncertainty margin in capital costs depending on the conditions of the storage site. A major downside of the PHS is its dependence to defined site conditions. Suitable sites are often located in mountainous regions that are often far apart from demand centers. This might be an opportunity for other storage technologies that are more flexible in this field. In the analysis the AA-CAES reaches closest to the PHS in day-storage operation. With some cost reduction this technology could become an alternative in the future.

Under current market conditions power-to-gas technologies such as H_2 and CH_4 are economically infeasible. With their low storage efficiency daily price-spreads are simply too small to generate sufficient revenues. However, the results show that they could increase their revenues considerably when operating as a long-term storage plant for weekly, monthly or even seasonal balancing. By feeding methane or to some extent hydrogen into the existing natural gas infrastructure they can access the vast natural gas storage capacities. This makes them the only power storage technologies with sufficient capacity to store huge quantities of excess power for long timer horizons in order to balance supply and demand between seasons. However, for a large-scale adoption of such plants they need to become a cost effective investment. Technological progress could lead to cost reductions in the future, but their profitability will also rely on the development of the power market prices.

In general the results show that revenues from load-levelling operation have decreased considerably in the analysed timer period. This indicates the strong impact of spot market prices on revenues of storage plants. There is evidence that the growth of PV capacity has been one of the main drivers of this development. Feed-in from PV has lowered price-peaks on summer days and thereby reduced storage revenues. Earnings from provision of ancillary services could partly compensate these losses, but their future development is uncertain, since ancillary service markets

Table A-1
Results for seasonal storage operation.

Year	Revenue (kW h/kW/year)	Max. capacity used (kW h/kW)	Power input (kW h/kW/year)	Derived maximum feasible investment costs (€/kW)
Day storage				
PHS				
2007	78	7	2920	904
2008	95	7	2909	1103
2009	58	7	2969	679
2010	44	7	2803	515
2011	38	7	2685	445
AA-CAES				
2007	60	7	2648	632
2008	64	7	2541	681
2009	40	7	2512	422
2010	25	7	2136	269
2011	19	7	1751	200
H_2				
2007	24	7	1440	259
2008	17	7	1008	176
2009	11	7	1119	119
2010	5	7	493	53
2011	3	7	298	31
CH_4				
2007	20	7	1226	210
2008	12	7	762	129
2009	8	7	889	89
2010	4	7	383	39
2011	2	7	215	22
Seasonal storage				
PHS				
2007	126	1373	4153	1473
2008	147	1056	3944	1707
2009	81	1062	3732	948
2010	62	887	3512	727
2011	54	572	3406	624
H_2				
2007	55	1226	2711	581
2008	35	733	1801	375
2009	23	698	1629	244
2010	9	452	876	99
2011	5	140	411	51
CH_4				
2007	44	1042	2340	463
2008	23	517	1355	248
2009	16	592	1328	173
2010	6	309	601	62
2011	3	108	277	33

in Europe are currently facing a consolidation process. All this has discouraged investors and many storage projects have been cancelled or postponed.

However, investing in storage plants requires a forward-looking perspective on the development of the power market. This would reveal, that the increased feed-in of intermittent RES-E could open new opportunities in the future. The expected further extension of PV capacity will create new load peaks, which can be utilized by storage plants. The planned extension of well-connected wind capacities all over Europe together with increasing carbon prices will drive many medium- and even base-load plants out of the market. This will reduce base load prices and steepen the merit order curve, leading to higher revenues from load levelling operation for power storage plants.

Extension of RES-E capacities will also drive demand for ancillary services, like provision of balancing energy or reserve capacity and reserve power, in the power market. An integrated European market for these services will provide a more stable market environment than national markets currently do. This will open new opportunities for power storage. A closer look on this issue will be subject to future research.

Appendix A. Detailed results for seasonal storage

See Table A-1.

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